

Software System Design (2025) - Architecture

Assignment 1

(20 Points)

You are responsible to INDEPENDENTLY complete three individual tasks related to software quality attributes and tactics, then compile and submit your report.

DEADLINE: 12:00am on (Friday) May 16, 2025

Task 1: Quality Attribute Scenarios (8 points)

With the rapid development of Artificial Intelligence (AI), it has emerged as an enabling component in more and more contemporary software systems. Do your own research on AI-Native Applications based on AI Agents, then figure out **FOUR quality attributes** with the highest priority that have to be seriously considered in the architecture of such applications, at least **TWO of which are NOT introduced in lectures**. Apply scenario-based (stimulus-response) analysis method to create their **general scenarios** and typical **concrete scenarios** (using six-element analysis with at least **THREE response measures**) respectively based on your research and understanding. Present the general scenario in the format of six-element table and the concrete scenario in the stimulus-response diagram.

Task 2: Tactics (6 points)

Discuss at least **TWO** possible **strategies** to improve each quality attribute; propose at least **TWO** different **tactics** for each possible strategy and identify their potential **impacts on other attributes**. Use a table with columns of quality attributes and rows of strategies/tactics to discuss the benefits and the penalties of the potential impacts that may come with each tactic.

Task 3: Design Decisions (6 points)

Follow the seven categories of architectural design decisions, enumerate at least **TWO** most important **design decisions** (in your point of view) under **each category** respectively for the **TWO quality attributes NOT introduced in lectures**. Elaborate the **rationale** supporting each of your proposed design decisions.

Assignment Deliverable:

The assignment deliverables should be submitted individually. Each deliverable contains no more than 10 pages (A4 size) plus a cover page. Note your individual submission will be plagiarism and AI-generated content checked. You'll receive no point if plagiarism is detected in any task of your assignment.